Honors Calculus III, Second midterm.

This exam consists of two problems for a total of 30 points. Attempt to solve as many problems as you can. Make sure you read each question carefully and understand what you are being asked to do. You should progress sequentially, justify each step, use notation correctly, and clearly mark the final answer. Please write your name and section number on every answer booklet you use.

In this exam you can use results from lectures or homework, or Math 16100/16200, provided you state them clearly and you avoid circular arguments: for example, you cannot use a result that you are explicitly asked to prove in the question. As usual, for full credit you need to show and explain your work.

Problem 1. Let $f : \mathbf{R}^2 \to \mathbf{R}$ be a function. We define

$$f^{[1]} : \mathbf{R} \to \mathbf{R}, f^{[1]}(x) = f(x, 0)$$
$$f^{[2]} : \mathbf{R} \to \mathbf{R}, f^{[2]}(x) = f(0, x).$$

1. (3pt) Define what it means for f to be continuous.
2. (4pt each) Prove or disprove the following assertions:
   - If f is continuous, then $f^{[1]}$ is continuous.
   - If $f^{[1]}$ and $f^{[2]}$ are continuous, then f is continuous.
   - If $f^{[1]}$ and $f^{[2]}$ are linear, then f is linear.

Problem 2.

1. (3pt each) Let $X \subset \mathbf{R}^n$. Define what it means for X to be open, and define what it means for X to be compact.
2. (3pt each) Determine whether the following sets are open, or closed, or neither, or both.

$$X = \{(x, y, z) \in \mathbf{R}^3 : x^3 + y^2 - z^5 \leq 1\}$$
$$Y = \{(x, y, z) \in \mathbf{R}^3 : 0 < x^2 + y^2 + z^2 < 1\}$$
$$Z = \{(x, y, z \in \mathbf{R}^3 : 2xz - x^2 - z^2 = 1\}.$$

Solutions to Problem 1.

1. If $v \in \mathbf{R}^2$, we say that f is continuous at v if for all $\epsilon > 0$ there exists $\delta > 0$ such that $w \in \mathbf{R}^2$ and $||v-w|| < \delta$ implies $|f(v)-f(w)| < \epsilon$. We say that f is continuous if f is continuous at v for all $v \in \mathbf{R}^2$.

2. The first assertion is true, since the function $\iota_1 : \mathbf{R} \to \mathbf{R}^2, x \mapsto (x, 0)$ is continuous, and $f^{[1]} = f \circ \iota_1$, and the composite of two continuous functions is continuous.

3. The second assertion is false (see HW4). For an explicit counterexample, let $f(x, y) = 1$ if $xy = 0$ and 0 otherwise, so that f is the indicator function of the union of the coordinate axes. Then $f^{[1]}$ and $f^{[2]}$ are constant functions, and so they are continuous. However, $\lim_{n\to\infty}(1/n, 1/n) = (0, 0)$ but $\lim_{n\to\infty} f(1/n, 1/n) = 0$ and $f(0, 0) = 1$, hence f is not continuous at $(0, 0)$ by a result from HW4.

4. The third assertion is also false: any nonlinear function f such that $f^{[1]} = f^{[2]} = 0$ is a counterexample. For instance, you can take $f(x, y) = 0$ if $xy = 0$ and 1 otherwise.

Solutions to Problem 2.

1. We say that X if open if for all $x \in X$ there exists an open rectangle $A \subset \mathbf{R}^n$ such that $x \in A$ and $A \subset X$. We say that X is compact if for every open cover $\{U_i\}_{i\in I}$ of X there exists a finite subset $J \subset I$ such that $X \subset \cup_{j\in J} U_j$.

2. X is closed, since it equals $g^{-1}(-\infty, +1]$ for the continuous function $g : \mathbf{R}^3 \to \mathbf{R}, g(x, y, z) = x^3+y^2-z^5$. Since X is not empty (it contains $(0, 0, 0)$) and X is not equal to $\mathbf{R}^3$ (it does not contain $(2, 0, 0)$) it follows from a result from class that X cannot be both open and closed, and so X is not open.

3. Y is open, because it equals $h^{-1}(0, 1)$ for the continuous function $h(x, y, z) = x^2 + y^2 + z^2$. By the same argument as before, it is not closed; we can also check this directly, since $(1/n, 0, 0) \in Y$ for all $n > 0$ but $\lim_{n\to\infty}(1/n, 0, 0) \notin Y$.

4. We have $(x, y, z) \in Z$ if and only if $(x - z)^2 = -1$, hence Z is empty, and so Z is open and closed.